

PAPER → CODE · ARXIV:1810.04805

BERT

Bidirectional

Transformers

Every equation of Devlin, Chang, Lee & Toutanova (Google AI, 2018), explained parameter by parameter — the deleted causal mask, the 15% / 80-10-10 masked-language-model objective, NSP, the input recipe, and the fine-tuning heads that set 11 states of the art.

The math explained — companion to the interactive site

Original authors: J. Devlin, M.-W. Chang, K. Lee, K. Toutanova

Visualised & annotated by **Abdulsamad Teniola Muyideen**

7 core equations · Base/Large configs · runnable char-level twin at
Nanogpt/nanobert.py

The one-line difference

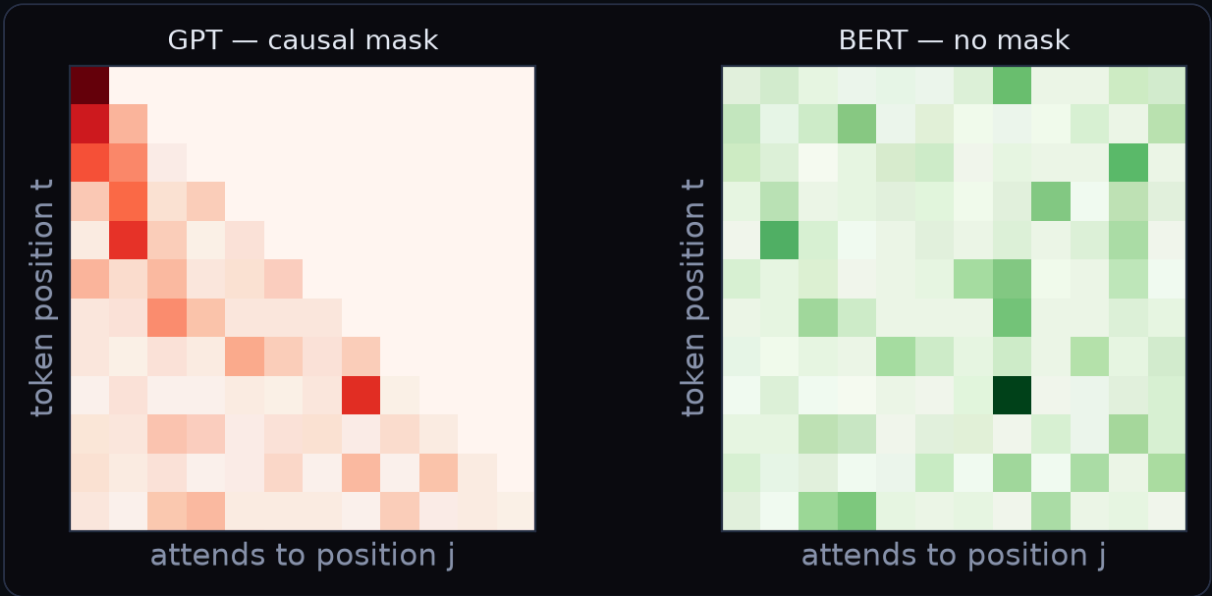
SELF-ATTENTION, SHARED BY GPT AND BERT

(att)

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right) V$$

What it does. Every token forms a query Q , a key K and a value V ; the scaled dot-products decide how much each position reads from every other. The entire GPT/BERT architectural split is the mask M : GPT sets $M = -\infty$ above the diagonal so no token sees its future; BERT sets $M = 0$ — the mask is deleted and every token attends in both directions. That deletion is why BERT represents language better, and why it cannot generate text: it was never trained to predict without seeing the right-hand context.

SYMBOL	MEANING	VALUE
Q, K, V	learned linear projections of the input	—
$\sqrt{d_k}$	scaling that keeps softmax gradients healthy	—
$M(\text{GPT})$	$-\infty$ above the diagonal — causal	autoregressive
$M(\text{BERT})$	all zeros — bidirectional	understanding



The same random attention scores under the two masks: GPT's world is triangular; BERT's is the full square.

Masked language modelling

THE OBJECTIVE

(MLM)

$$\mathcal{L}_{\text{MLM}} = - \sum_{i \in \mathcal{M}} \log P(x_i \mid \tilde{x}), \quad |\mathcal{M}| \approx 0.15 T$$

What it does. Bidirectional attention makes next-word prediction cheatable (each position could look the answer up), so BERT invents a new exam: corrupt the input x into $\tilde{x}$, then reconstruct the ORIGINAL tokens at the corrupted 15% of positions only. The loss is ordinary cross-entropy, restricted to the set $\mathcal{M}$ of selected positions.

SYMBOL	MEANING	VALUE
$\mathcal{M}$	the ~15% selected positions	15%
$\tilde{x}$	the corrupted input sequence	—

THE CORRUPTION RECIPE

(80/10/10)

for $i \in \mathcal{M}$: 80% $\rightarrow$ [MASK], 10% $\rightarrow$ random token, 10% $\rightarrow$ unchanged

What it does. Why not replace all selected tokens with [MASK]? Because [MASK] never appears at fine-tuning or inference time — a model trained 100% on it would learn that only masked slots carry gradient. The 10% random substitutions and 10% left-unchanged (but still graded) force the encoder to maintain an honest contextual representation of every position, since it never knows which ones will be examined.

SYMBOL	MEANING	VALUE
80	replaced by [MASK]	—
10	replaced by a random vocabulary token	—
10	left as-is, still predicted	—

NSP and the input recipe

NEXT SENTENCE PREDICTION

(NSP)

$$\mathcal{L}_{\text{NSP}} = -\log P(\text{IsNext} \mid \mathbf{C}), \quad \mathbf{C} = h_{[\text{CLS}]}$$

What it does. 50% of training pairs are consecutive sentences, 50% random; a binary classifier on the final [CLS] vector must tell them apart — intended to teach sentence-level relations for QA and NLI. History's verdict: the paper's own ablations show modest gains and RoBERTa (2019) dropped NSP entirely without loss. The MLM objective is the part of BERT that mattered.

SYMBOL	MEANING	VALUE
C	final hidden state of [CLS]	sequence summary

THE INPUT REPRESENTATION

(input)

$$\text{input}_i = E_{\text{tok}}(x_i) + E_{\text{seg}}(A/B) + E_{\text{pos}}(i)$$

What it does. Three embeddings summed per token: WordPiece identity (30,000-piece subword vocabulary), a segment flag marking sentence A or B, and a learned position embedding. Special tokens [CLS] (front, sequence-level summary) and [SEP] (sentence boundaries) give every downstream task the same plug-in shape.

SYMBOL	MEANING	VALUE
<i>WordPiece</i>	subword vocabulary	30k
$[CLS]/[SEP]$	summary slot / boundary marker	—

Architecture & pre-training

PARAMETER BUDGET

(size)

$$N_{\text{non-emb}} \approx 12 L H^2 \quad \text{Base: } 12 \cdot 12 \cdot 768^2 \approx 85\text{M} \text{ (+ emb } \approx 110\text{M)}$$

What it does. BERT-Base: L=12 layers, H=768, A=12 heads, feed-forward 4H — 110M parameters, deliberately sized to match GPT-1 for a fair comparison. BERT-Large: L=24, H=1024, A=16 — 340M, the configuration behind the headline results. Pre-trained on BooksCorpus + English Wikipedia (3.3B words) for 40 epochs with GELU activations and Adam. In post-Chinchilla hindsight (~30 tokens/param), Base was accidentally near compute-optimal.

SYMBOL	MEANING	VALUE
<i>Base</i>	$L_{12} \cdot H_{768} \cdot A_{12}$	110M params
<i>Large</i>	$L_{24} \cdot H_{1024} \cdot A_{16}$	340M params
<i>data</i>	BooksCorpus + Wikipedia	3.3B words

Fine-tuning heads

CLASSIFICATION HEAD (GLUE)

(cls)

$$P(y \mid \text{sent}) = \text{softmax}(W \mathbf{C}), \quad W \in \mathbb{R}^{K \times H}$$

What it does. Sentiment, entailment, similarity: one new $K \times H$ matrix on the [CLS] vector, then fine-tune everything for 2–4 epochs. A few thousand new parameters steering 110M pre-trained ones. GLUE at publication: 80.5%, +7.7 points over the prior state of the art.

SYMBOL	MEANING	VALUE
W	the only new parameters	$K \cdot H$

SPAN HEAD (SQUAD)

(span)

$$P_{\text{start}}(i) = \frac{e^{S \cdot h_i}}{\sum_j e^{S \cdot h_j}}, \quad P_{\text{end}}(i) = \frac{e^{E \cdot h_i}}{\sum_j e^{E \cdot h_j}}, \quad (\hat{j}, \hat{k}) = \arg \max_{j \leq k} S \cdot h_j + E \cdot h_k$$

What it does. Extractive QA: the answer is a span of the passage. Two learned vectors S and E score every token as a candidate start/end; the best valid pair (start before end) wins. Only $2 \cdot H$ new parameters — 93.2 F1 on SQuAD v1.1, surpassing human performance. The pattern across all heads: bidirectional pre-training did the hard work; the task head is nearly free. The key ablation: the same model with a left-to-right mask drops from 84.9 to 77.8 F1 — bidirectionality is the ingredient.

SYMBOL	MEANING	VALUE
S, E	start / end scoring vectors	$2 \cdot H$ params
<i>ablation</i>	LTR (causal) vs bidirectional on SQuAD	77.8 vs 84.9 F1